

Geometry

Unit 9

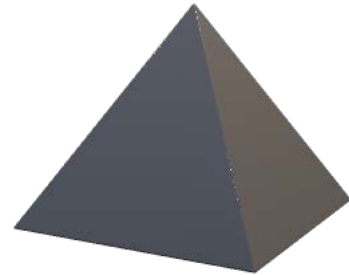
Lesson 5 Homework

Name_____ **Answer Key** _____

Date_____

Period_____

Using the given three figures, answer questions 1-3.



1. Which two figures can have the same volume?

The tube and the cylinder

2. Explain how the figures listed in Question 1 can have the same volume.

Explanations may vary, but one explanation is that the cross-sections of the pyramid can't be equal in area to the cross-sections of the other two.

3. What information is required for each figure to determine if the volumes are equal?

Radius and height of each

4. Isaac and Dallas went to an amusement park. They each bought a cheeseburger. Isaac's meat patty came with a square base. Dallas's meat patty had a circular base. Is it possible that they both purchased the same size of cheeseburger?

Yes, if the cross-sections have the same area.



5. Dallas's meat patty was 0.75 inches high with a radius of 4.5 inches. What is the volume?

$\pi(4.5)^2(0.75) \approx 47.69 \text{ in}^3$

6. Isaac's hamburger patty was also 0.75 inches high but with a base whose side length was 4.5 inches. Determine if the two containers have the same volume.

$$0.75(4.5)^2 \approx 15.19 \text{ in}^3$$

Not the same volume

7. Brand A nail polish is sold in rectangular prism jars while Brand B nail polish is sold in cylinder jars. Is it possible for a given quantity of Brand A nail polish and Brand B nail polish to have the same volume?

Yes, if the cross-sections have the same area.



8. A jar of Brand B nail polish is 3 inches tall, with a diameter of 1.25 inches. In a store display, there are a total of 18 jars. What is the volume of nail polish in the display?

$$18 \cdot 3(0.625)^2\pi \approx 66.23 \text{ in}^3$$

9. One jar of Brand A nail polish is 3 inches tall, with a width of 1 inch and a length of 1.25 inches. There are 18 jars in the display. What is the total volume of nail polish in the display?

$$18(3)(1)(1.25) = 67.5 \text{ in}^3$$

10. Compare the answers from Questions 8 and 9.

Brand A has the larger volume.